

Practical 1

Goal: To familiarise yourself with OrCAD Schematic Capture and Pspice, which will be used to simulate DC circuits.

Tasks:

Practical 1 consists of three parts:

1. A pre-practical assignment that must be completed individually in the week that precedes practical 1.
2. A pre-practical test that must be completed at the start of the practical session.
3. The practical itself that will be completed in groups of two during the practical session.

TASK 1

In order to complete the assignment, you will need to use the OrCAD software. A tutorial has been created that you can use to both install the software and familiarise yourself with the program's functionality.

The tutorial can be downloaded from here:

http://www.ee.up.ac.za/main/_media/en/undergrad/subjects/ebn122/dab_a_first-year_s_guide_to_installing_orcad_on_your_home_pc_and_designing_your_first_circuit.pdf

Be sure to complete this tutorial before starting with the assignment.

The pre-practical assignment can be found on EBN111's clickUP page.

This assignment will be active for the entire week; so you do not have to complete the practical in one sitting. Make sure to save your answers as you go along so that you do not lose your answers.

Start the pre-practical assignment when you are ready and submit your assignment before the end of the week.

TASK 2

The pre-practical test will be written at the start of your practical session and the questions will test you in the following areas:

- a) Information provided during the pre-practical lecture
- b) Information contained in the slides provided for practical 1
- c) Information obtained from completing the tutorial
- d) Information obtained from completing the pre-practical assignment itself.

The test will be answered individually and will take up 30 minutes of the practical session. After completing the test, you will move on to the practical itself.

TASK 3

The practical itself must be completed during the practical session. Students may work in groups of 2.

Print out the Practical 1 Guidelines (found on ClickUP) and bring this to the practical session. **DO NOT FORGET TO PRINT THIS AND BRING IT ALONG!**

The guidelines will make the practical a lot easier and will provide vital information during the practical.

Follow the instructions, perform the simulations and then fill in your answers.

After you have completed the practical and have filled in all the values, go onto ClickUP and complete the Practical 1 Evaluation.

NOTE THAT THIS IS THE PART THAT COUNTS MARKS AND MUST BE COMPLETED DURING THE PRACTICAL WEEK!

In conclusion, you therefore have to submit the following ClickUP evaluations in order to pass the practical:

- 1) Pre practical Assignment
- 2) Practical test
- 3) Practical 1 evaluation.